

Harish Lal

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Experience

Senior Product Development Engineer - UI, Envestnet Asset Management Ltd Aug 2023 – Present

- Consolidated **15 grid implementations** into a reusable grid system and helped scale a **unified widget framework supporting 80+ widgets**, reducing duplicated UI logic by **~60%**.
- Introduced **config-driven widgets and grids** enabling product teams to update metrics, layouts, and configurations without engineering changes, significantly accelerating feature rollout.
- Co-developed a **Custom Layout feature** allowing users to persist personalized **metric and grouping configurations** while preserving system defaults, improving workflow efficiency by **~35%**.
- Led development of the **Metric Panel within the Grid Editor**, enabling dynamic metric selection and column configuration; refactored a legacy grouping module from **~1000+ lines to <400 lines** using reusable hooks.
- Enhanced the existing **“See More” functionality** by building a **slide-out panel interface** that surfaces filtered grid data within the current page, enabling scalable interactions across **150+ widgets**.
- Awarded **Star of the Quarter** for strong ownership and platform refactoring initiatives that removed **1.5k+ lines of redundant code** and improved modularity and maintainability.

Projects

ArchFlow – Visual Distributed System Simulator Next.js, TypeScript, React Flow, Zustand

- Created a **drag-and-drop architecture modeling platform** supporting **6 infrastructure node types** (API gateway, service, cache, queue, worker, database) for constructing distributed system workflows.
- Developed an **event-driven simulation engine** executing **1,000+ request flows per run** to emulate **latency propagation, queue buffering, and service failures** while generating real-time system metrics.
- Optimized graph rendering performance for **large architectures (50+ nodes, 120+ edges)** through **component memoization and state partitioning**, reducing unnecessary UI re-renders by **70%**.

Circle of Life – Location-Based Community Assistance Platform Go, PostgreSQL, Redis, Next.js

- Engineered a **location-aware community platform** enabling users to discover nearby posts using **geospatial filtering within configurable geographic radii**.
- Implemented **concurrent REST APIs in Go** supporting feed generation and messaging with **sub-150ms response latency** across **10K+ simulated posts**.
- Optimized **database schema and indexing strategy for geospatial queries**, improving feed query performance by **60% during load testing**.

Skills

Languages: Go, JavaScript, TypeScript, Python, C++, SQL

Frontend: React.js, Next.js, Redux, Tailwind CSS, HTML, CSS

Backend & Databases: PostgreSQL, Redis, REST APIs

Tools: Docker, Git, Postman

Education

Govt. Engg. College, Barton Hill, BTech in Electrical Engineering - **7.68 GPA** Aug 2019 – Jun 2023

- Orchestrated **two hackathons, over 10 technical events**, and a **web development workshop** for beginners with **100+ participants**, boosting interactive learning and practical skills as **Technical Head** of the Innovation and Entrepreneurship Development Cell (IEDC).

PM SHRI Kendriya Vidyalaya Pattom, Shift 2, AISSCE - Computer Science - **94.8%**

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